Uniqueness of the Internal Manifold: Deriving $\mathbb{C}P^2 \times S^3$ from Vacuum Axioms in Density Field Dynamics

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Abstract. We prove that $\mathbb{C}P^2 \times S^3$ is the unique internal manifold for the spectral completion of Density Field Dynamics (DFD), under six physically motivated axioms on the ψ-vacuum. The axioms encode chirality (an empirical fact about nature), multiplicative vacuum composition (a direct consequence of DFD's core postulate P1), ground-state stability, and minimality. They do not reference the Standard Model gauge group, fermion content, or any coupling constant. The gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, three generations, and $\alpha^{-1} = 137.036$ emerge as derived consequences. The product structure $K = K_C \times K_G$ is not assumed but forced by the logical incompatibility of the chirality requirement ($w_2 \neq 0$) with Lie-group parallelizability ($w_2 = 0$) on a single connected manifold.

1. Introduction

Density Field Dynamics [1] is a scalar refractive-index theory of gravity defined by two postulates:

P1. The optical refractive index of the vacuum is $n = e^{\psi}$, where $\psi(\mathbf{x}, t)$ is a scalar field on flat $\mathbb{R}^{3,1}$.

P2. Test bodies move under the conservative potential $\Phi = -c^2\psi/2$.

The DFD microsector—which derives $\alpha = 1/137$, the fermion mass spectrum, and the MOND interpolation function $\mu(x) = x/(1+x)$—requires a spectral completion on a product geometry $M = \mathbb{R}^{3,1} \times K$, where K is a compact internal manifold.

Previous versions of DFD postulated $K = \mathbb{C}P^2 \times S^3$. This paper derives it from six physically motivated axioms.

2. The Six Vacuum Axioms

Axiom 1 (Spectral completion)**.** The ψ-vacuum admits a UV completion via a spectral action

$$S_B = \mathrm{Tr}\, f(D^2/\Lambda^2) \tag{1}$$

on a product geometry $M = \mathbb{R}^{3,1} \times K$, where K is a compact Riemannian manifold, D is the Dirac operator on M, Λ is a UV cutoff, and f is a positive test function.

Motivation. The Chamseddine–Connes spectral action [2] is the canonical trace functional in noncommutative geometry that produces both the Einstein–Hilbert gravitational action and gauge kinetic terms from a single principle. DFD's postulates P1–P2 emerge as the weak-field limit of its a_4 Seeley–DeWitt coefficient.

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Axiom 2 (Kähler chirality)**.** At least one irreducible factor of K (in the de Rham decomposition) is a compact, simply-connected, Kähler–Einstein manifold with $w_2 \neq 0$ (spin^c but not spin).

Motivation. Nature violates parity [3]. Chiral fermions in 4D require a nonzero spin^c Dirac index on the internal space. The Hirzebruch–Riemann–Roch theorem—which computes this index as an integral of Chern characters—requires the Dirac operator to decompose as $\sqrt{2}(\bar{\partial} + \bar{\partial}^*)$. This decomposition takes its cleanest canonical form in the Kähler setting, where the spin^c Dirac operator and the Dolbeault operator coincide. The condition $w_2 \neq 0$ (spin^c-not-spin) ensures the index is non-trivially chiral: in the specific classification relevant here (compact simply-connected positive-curvature Kähler–Einstein manifolds of complex dimension ≤ 2), the spin candidates have vanishing chiral index, while the spin^c-not-spin candidate $\mathbb{C}P^2$ does not.

Axiom 3 (Topological composition)**.** At least one irreducible factor of K is a compact, simply-connected Lie group manifold.

Motivation. DFD's postulate P1 implies multiplicative composition of vacuum sectors: $n_{\text{total}} = n_1 \cdot n_2$. This composition is associative, has an identity ($n = 1$, i.e., $\psi = 0$), and every element has an inverse ($n \to 1/n$). In the spectral completion (V1), the internal degrees of freedom at each spacetime point form a fiber. For the vacuum composition law to be fiber-wise consistent—meaning that the composition of two vacuum configurations at the same spacetime point yields a valid vacuum configuration—the fiber must carry a group structure. A compact connected manifold admitting a continuous associative multiplication with identity and inverses is a topological group; since the underlying space is a finite-dimensional manifold, Hilbert's fifth theorem [11] implies that it is a Lie group. Simple-connectedness ensures the group is its own universal cover, so the composition has no topological ambiguity. V3 is the DFD-specific bridge from the multiplicative vacuum-composition law in P1 to an internal topological carrier of that composition. It is well-motivated within DFD but remains an axiom, not a tautology; in a different theory without multiplicative refractive composition, V3 would not apply.

Axiom 4 (Stability)**.** K is compact and simply-connected. Every irreducible factor in its de Rham decomposition is Einstein with positive Ricci curvature.

Motivation. The Einstein condition $\mathrm{Ric} = \lambda g$ ($\lambda > 0$) is the critical point of the spectral action at fixed volume—the vacuum ground state. A non-Einstein K would be unstable under Ricci flow and would relax to the Einstein point.

Axiom 5 (Minimality of dimension)**.** $\dim(K)$ is the minimum compatible with Axioms 1–4 and the existence of chiral zero modes (nonzero spin^c Dirac index).

Motivation. Each additional dimension introduces potential moduli that must be stabilized. The minimal-dimension vacuum has the cleanest low-energy spectrum.

Axiom 6 (Minimality of topology)**.** At fixed dimension, K has the minimum second Betti number $b_2(K)$ compatible with Axioms 1–5.

Motivation. Each independent 2-cycle supports an independent harmonic 2-form, hence an independent gauge modulus. Minimizing b_2 ensures the vacuum carries no unnecessary internal flux degrees of freedom.

Remark 2.1 (Axiom content). Axioms 1–6 do not reference the Standard Model gauge group, fermion representations, hypercharge assignments, or any coupling constant. Their physical sources are:

- V1: the spectral action framework—standard in noncommutative geometry [2];
- V2: parity violation—an empirical fact [3], combined with the mathematical requirement that the spin^c Dirac index factorizes cleanly (Kähler condition);
- V3: fiber-wise consistency of the multiplicative vacuum-composition law $n = e^{\psi}$—a direct consequence of DFD's postulate P1 combined with the spectral completion V1;
- V4: vacuum ground-state stability—a standard physical requirement;
- V5, V6: parsimony applied to dimension and topology—standard in physics.

3. Derivation

3.1. Step 0: Product structure is forced.

Lemma 3.1 (Forced product decomposition)**.** *The factors required by Axioms 2 and 3 must be distinct irreducible components of K. In particular, K is not irreducible.*

Proof. All compact simply-connected Lie groups are parallelizable: they admit a global frame of left-invariant vector fields, so $TG \cong G \times \mathbb{R}^n$ is trivial. Therefore $w_1(G) = w_2(G) = 0$ for any compact simply-connected Lie group G.

Axiom 2 requires $w_2 \neq 0$ on the Kähler–chirality factor. Since $w_2 = 0$ on any Lie group factor, the Kähler–chirality factor cannot be the Lie group factor. They are distinct irreducible components.

By Axiom 4, K is compact and simply-connected. The de Rham decomposition theorem [9] guarantees that K splits uniquely (up to order) as a Riemannian product of irreducible factors:

$$K = K_1 \times K_2 \times \cdots \times K_r. \tag{2}$$

At least one factor (call it K_C) satisfies Axiom 2, and at least one distinct factor (call it K_G) satisfies Axiom 3. □

3.2. Step 1: The chirality factor is $\mathbb{C}P^2$.

Lemma 3.2 (Classification of Kähler–chirality factors)**.** *Let X be a compact, simply-connected, Kähler–Einstein manifold with positive scalar curvature and $w_2(X) \neq 0$. Then $\dim_{\mathbb{R}}(X) \geq 4$, and the unique such manifold of minimal real dimension with $b_2 = 1$ is $X = \mathbb{C}P^2$.*

Proof. Real dimension 2 (complex dimension 1). The compact simply-connected Kähler 1-folds are the Riemann surfaces of genus 0, i.e., $\mathbb{C}P^1 \cong S^2$. But S^2 is spin ($w_2(S^2) = 0$), so it violates $w_2 \neq 0$. Excluded.

Real dimension 4 (complex dimension 2). The compact simply-connected Kähler–Einstein surfaces with $c_1 > 0$ are classified. By Tian [4] and Tian–Yau [5], the del Pezzo surfaces admitting Kähler–Einstein metrics are determined by the Matsushima–Lichnerowicz obstruction [6, 7] (the automorphism group must be reductive) and the Tian–Yau α-invariant:

Surface	b_2	KE metric?	w_2	Status
$\mathbb{C}P^2$	1	✓ (Fubini–Study)	$H \neq 0$	**Admitted**
$\mathbb{C}P^1 \times \mathbb{C}P^1$	2	✓ (product)	0	Spin; excluded by $w_2 \neq 0$
$\mathbb{C}P^2 \# \overline{\mathbb{C}P^2}$	2	× (Matsushima)	$\neq 0$	**No KE metric exists**
$\mathbb{C}P^2 \# 2\overline{\mathbb{C}P^2}$	3	× (Matsushima)	$\neq 0$	No KE metric exists
$\mathbb{C}P^2 \# k\overline{\mathbb{C}P^2}$, $k = 3, \ldots, 8$	$k{+}1$	✓ (Tian–Yau)	$\neq 0$	Excluded by V6 ($b_2 > 1$)

The critical exclusion: $\mathbb{C}P^2 \# \overline{\mathbb{C}P^2}$ **does not admit a Kähler–Einstein metric.** This is the Matsushima–Lichnerowicz theorem [6, 7]: a necessary condition for a compact Kähler manifold to admit a Kähler–Einstein metric with $c_1 > 0$ is that its automorphism group be reductive. The automorphism group $\mathrm{Aut}(F_1)$ of $F_1 = \mathbb{C}P^2 \# \overline{\mathbb{C}P^2}$ (the first Hirzebruch surface) contains a unipotent radical and is therefore not reductive. No Kähler–Einstein metric exists.

Remark 3.3 (Page metric exclusion). The Page metric [10] on $\mathbb{C}P^2 \# \overline{\mathbb{C}P^2}$ is Einstein but not Kähler. Axiom 2 requires Kähler–Einstein, so the Page metric is excluded at the axiom level, not by post hoc selection.

The del Pezzo surfaces $\mathbb{C}P^2 \# k\overline{\mathbb{C}P^2}$ for $k = 3, \ldots, 8$ do admit Kähler–Einstein metrics [4], but they have $b_2 = k + 1 \geq 4$, and are excluded by Axiom 6: $\mathbb{C}P^2$ has $b_2 = 1$, which is strictly less.

Therefore $\mathbb{C}P^2$ is the unique manifold satisfying Axiom 2 in real dimension 4 with minimal b_2. □

Corollary 3.4. *The Kähler–chirality factor in K is $K_C = \mathbb{C}P^2$* (dim = 4).

3.3. Step 2: The group factor is S^3.

Lemma 3.5 (Classification of minimal Lie group factors)**.** *The unique compact simply-connected Lie group of minimal positive dimension is* $\mathrm{SU}(2) \cong S^3$ (dim = 3).

Proof. The compact simply-connected Lie groups are classified by Cartan. Their dimensions begin:

Group	Rank	Dimension
SU(2)	1	3
SU(3)	2	8
$\mathrm{Sp}(4) \cong \mathrm{Spin}(5)$	2	10
G_2	2	14

No compact simply-connected Lie group exists in dimension 1 ($\mathrm{U}(1) \cong S^1$ is not simply-connected) or dimension 2 (no Lie algebra of dimension 2 is both semisimple and compact; the only 2-dimensional Lie algebras are the abelian $\mathbb{R}^2$ and the solvable $\mathrm{aff}(1)$, neither yielding a compact simply-connected group). Therefore $\mathrm{SU}(2) \cong S^3$ is the unique compact simply-connected Lie group of minimum positive dimension. □

Corollary 3.6. *By Axioms 3 and 5, the Lie group factor is $K_G = S^3$* (dim = 3).

3.4. **Step 3:** $K = \mathbb{C}P^2 \times S^3$ **and nothing else.**

Theorem 3.7 (Uniqueness of the internal manifold)**.** *Under Axioms 1–6, $K = \mathbb{C}P^2 \times S^3$.*

Proof. By Lemma 3.1, K has at least two irreducible factors: K_C (satisfying Axiom 2) and K_G (satisfying Axiom 3).

By Corollary 3.4, $K_C = \mathbb{C}P^2$ (dimension 4).

By Corollary 3.6, $K_G = S^3$ (dimension 3).

By Axiom 5 (minimize $\dim(K)$):

$$\dim(K) \geq \dim(K_C) + \dim(K_G) = 4 + 3 = 7. \tag{3}$$

Any additional irreducible factor K_3 would satisfy $\dim(K_3) \geq 1$, giving $\dim(K) \geq 8 > 7$, violating minimality.

Could the chirality factor have $\dim > 4$? The next Kähler–Einstein spin^c-not-spin option is $\mathbb{C}P^3$ ($\dim = 6$), giving $\dim(K) \geq 6 + 3 = 9 > 7$. Excluded by Axiom 5.

Could the Lie group factor have $\dim > 3$? The next option is SU(3) ($\dim = 8$), giving $\dim(K) \geq 4 + 8 = 12 > 7$. Excluded by Axiom 5.

Could a non-product 7-manifold satisfy all axioms? No. Axiom 3 requires a Lie group factor; Axiom 2 requires a Kähler factor with $w_2 \neq 0$. All compact simply-connected Lie groups are parallelizable, hence $w_2 = 0$. These two requirements are incompatible on a single connected manifold. The product decomposition $K = K_C \times K_G$ is forced (Lemma 3.1).

Verification of all axioms on $K = \mathbb{C}P^2 \times S^3$:

- V1: $\mathbb{C}P^2 \times S^3$ is compact, Riemannian, spin^c; the spectral action is well-defined. ✓
- V2: $\mathbb{C}P^2$ is Kähler–Einstein (Fubini–Study, $\mathrm{Ric} = 6g$), with $w_2 = H \neq 0$. ✓
- V3: $S^3 \cong \mathrm{SU}(2)$ is a compact simply-connected Lie group. ✓
- V4: $\mathbb{C}P^2$ is Einstein ($\mathrm{Ric} = 6g$), S^3 is Einstein ($\mathrm{Ric} = 2g$), both with positive Ricci curvature. K is simply-connected (product of simply-connected spaces). ✓
- V5: $\dim(K) = 7$, the minimum possible. ✓
- V6: $b_2(K) = b_2(\mathbb{C}P^2) + b_2(S^3) = 1 + 0 = 1$, the minimum possible for any K satisfying V1–V5. ✓

Chiral zero-mode existence (consistency check). The spin^c Dirac index on $\mathbb{C}P^2 \times S^3$ is nonzero. By Künneth factorization [1, Theorem F.9], the index factors over the product. With the canonical spin^c structure on $\mathbb{C}P^2$ and minimal-energy flux configuration on S^3, the index equals 3, confirming chiral zero modes exist. This is a consistency check on the derived manifold, not an input to the selection.

Therefore $K = \mathbb{C}P^2 \times S^3$ is the unique solution to Axioms 1–6. □

4. What Emerges

Once $K = \mathbb{C}P^2 \times S^3$ is derived rather than postulated, the entire DFD microsector follows with zero continuous free parameters:

Output	Mechanism	Ref. in [1]
SU(3)×SU(2)×U(1)	Minimal $(3, 2, 1)$ partition	Prop. XVII.1
$N_{\text{gen}} = 3$	$\lvert k_3 \cdot k_2 \cdot q_1 \rvert = \lvert 1{\cdot}1{\cdot}3 \rvert$	Thm. F.13
$\alpha^{-1} = 137.036$	Chern–Simons at $k_{\max} = 60$	Thm. K.1
$\mu(x) = x/(1{+}x)$	S^3 composition law	Thm. N.8
$a_0 = 2\sqrt{\alpha}\, cH_0$	Scaling stationarity	Thm. N.14
$\bar{\theta} = 0$ (Strong CP)	Mapping torus dim = 8 (even)	Thm. L.3
Proton stability	$\pi_3(S^3) = \mathbb{Z}$ winding	Thm. F.17
9 fermion masses	Spin^c exponents + A_5 prefactors	App. K

5. Honest Assessment of Axiom Content

The six axioms do not reference the Standard Model, but they are not content-free:

- **V2 imports an empirical fact:** parity violation. This is observational input, not SM-theoretical input. Any vacuum theory must accommodate it. The Kähler requirement within V2 is motivated by the spectral action framework (V1): the index computation factorizes cleanly in the Kähler setting.
- **V3 is DFD-specific:** V3 bridges the multiplicative vacuum-composition law in P1 to an internal topological carrier of that composition via the fiber-wise consistency requirement of the spectral completion (V1). It is not a universal physical requirement—it is specific to theories with $n = e^{\psi}$ multiplicative refractive composition. Within DFD, it is well-motivated but remains an axiom, not a tautology.
- **V6 serves as a tiebreaker:** V6 is invoked only to exclude del Pezzo surfaces with $k \geq 3$ that admit Kähler–Einstein metrics. Without V6, $\mathbb{C}P^2$ would still be the preferred chirality factor by V5 (all competitors have the same dimension but more topology), but V6 makes the exclusion explicit rather than implicit.

The derivation is not assumption-free. It is assumption-*minimal*: six qualitative physical requirements produce a unique topological answer from which 30+ quantitative predictions follow.

6. Conclusion

Theorem (Uniqueness of the internal manifold)**.** *Under Axioms V1–V6, the internal manifold in the spectral completion of DFD is uniquely $K = \mathbb{C}P^2 \times S^3$.*

The key mathematical ingredients are:

(1) **Incompatibility lemma:** Lie groups are parallelizable ($w_2 = 0$), so V2 ($w_2 \neq 0$) and V3 (Lie group) force a product decomposition. This is derived, not assumed.
(2) **Matsushima–Lichnerowicz obstruction:** $\mathbb{C}P^2 \# \overline{\mathbb{C}P^2}$ does not admit a Kähler–Einstein metric, killing the only non-trivial competitor to $\mathbb{C}P^2$ in dimension 4. This is a classical theorem [6], not a new postulate.
(3) **Cartan classification:** SU(2) $\cong S^3$ is the unique compact simply-connected Lie group of minimum positive dimension. This is textbook.
(4) **Dimensional arithmetic:** $4 + 3 = 7$ leaves no room for additional factors.

Section XVIII.B.d of [1] stated: *"The only genuinely open theoretical question is the origin of the $\mathbb{C}P^2 \times S^3$ topology itself."*

Under V1–V6, the answer is: six physical requirements on the ψ-vacuum—spectral completion, Kähler chirality, multiplicative composition, ground-state stability, and two minimality conditions—admit exactly one solution. Whether those six axioms are themselves derivable from deeper principles is a separate question; what this paper establishes is that they suffice to uniquely determine K, eliminating the postulational status of the internal manifold.

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